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Conference support system and conference support method

Abstract

Provided is a conference support system that includes a user information acquisition unit that acquires voice information of each of users, and an analysis unit that generates text data of each user statement on the basis of the voice information, wherein the user information acquisition unit acquires classification information of a problem proposal or problem handling assigned by the users to the text data of the user statement, and acquires classification information of a satisfaction level or a problem resolution level assigned by the users to the text data to which the classification information of the problem proposal or the problem handling is assigned.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is a U.S. National Phase of International Patent Application No. PCT/JP2020/023032 filed on Jun. 11, 2020. Each of the above-referenced applications is hereby incorporated herein by reference in its entirety.

FIELD

(2) The present disclosure relates to a conference support system, a conference support method, and a conference support program.

BACKGROUND

(3) As a system that supports a conference held by a plurality of users, there are a system that visualizes biological information of users participating in the conference at the time of proceeding of the conference and proposes appropriate conference proceeding timing to alleviate stress of the users, and a technology that proposes how the users should make statements (see Patent Literature 1).

CITATION LIST

Patent Literature

(4) Patent Literature 1: JP 2019-191977 A

SUMMARY

Technical Problem

(5) Here, it is difficult to support proposal of a solution to a problem that is determined to be solved by a conference body and that is a reason for formation of the conference body. In addition, there is a case where it becomes necessary to change a problem itself set in a conference body since problem setting of the conference body is not appropriate, and there is a case where a solution is not presented to a problem that members of a conference body have overlooked. However, in the above-described technology, it is not considered to predict and propose a problem generated in the future during progress of a conference, or to predict and propose a problem overlooked by members of a conference body. In addition, it is difficult to improve accuracy by a proposal based only on biological information and context, and it is difficult to highly efficiently output a proposal adopted by users.

(6) Thus, there is a demand for a technology that predicts a problem to be solved by a conference body in the future and a problem overlooked by users participating in the conference body according to progress of a conference and that accurately gives proposal according to psychological states of the users.

(7) The present disclosure proposes a conference support system, conference support method, and conference support program capable of predicting a problem to be solved by a conference body in the future, a problem overlooked by members of the conference body, and the like according to progress of a conference, and of accurately giving proposal according to psychological states of users.

Solution to Problem

(8) According to the present disclosure, a conference support system includes: a user information acquisition unit that acquires voice information of each of users; and an analysis unit that generates text data of each user statement on a basis of the voice information, wherein the user information acquisition unit acquires classification information of a problem proposal or problem handling assigned by the users to the text data of the user statement, and acquires classification information of a satisfaction level or a problem resolution level assigned by the users to the text data to which the classification information of the problem proposal or the problem handling is assigned.

(9) Moreover, according to the present disclosure, a conference support method includes the steps of: acquiring voice information of a user; converting a statement of the user from the voice information into text data; acquiring classification information with respect to the text data; learning relevance between the classification information and the text data and generating a first learned model; and automatically assigning the classification information to the text data by using the first learned model.

(10) Moreover, according to the present disclosure, a conference support program causes a computer to execute the steps of: acquiring voice information of a user, converting a statement of the user from the voice information into text data, acquiring classification information with respect

to the text data, learning relevance between the classification information and the text data and generating a first learned model, and automatically assigning the classification information to the text data by using the first learned model.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. **1** is a view illustrating a configuration example of a conference support system according to an embodiment of the present disclosure.

(2) FIG. **2** is a view illustrating a configuration example of an information processing device according to the embodiment of the present disclosure.

(3) FIG. **3** is a view illustrating a configuration example of a user information processing unit according to the present disclosure.

(4) FIG. **4** is a view illustrating a configuration example of a conference progress prediction unit according to the present disclosure.

(5) FIG. **5** is a view illustrating a configuration example of minutes data according to the present disclosure.

(6) FIG. **6** is a schematic diagram of a conference progress status display displayed on an output unit of the information processing device.

(7) FIG. **7** is a flowchart for describing a procedure of information processing by a natural language processing unit.

(8) FIG. **8** is a description diagram for describing definitions of a “successful conference” and an “unsuccessful conference”.

(9) FIG. **9** is a flowchart illustrating a procedure of information processing by the conference progress prediction unit according to the present disclosure.

(10) FIG. **10** is a flowchart illustrating a procedure of information processing by a proposal unit.

(11) FIG. **11** is a description diagram for describing a function of the proposal unit.

(12) FIG. **12** is a flowchart illustrating information processing by the proposal unit according to the present disclosure.

(13) FIG. **13** is a description diagram for describing the function of the proposal unit.

DESCRIPTION OF EMBODIMENTS

(14) In the following, embodiments of the present disclosure will be described in detail on the basis of the drawings. Note that in each of the following embodiments, overlapped description is omitted by assignment of the same reference sign to the same parts.

(15) FIG. **1** is a view illustrating a configuration example of a conference support system according to an embodiment of the present disclosure. As illustrated in FIG. **1**, a conference support system **1** includes three information processing devices **10**A, **10**B, and **10**C, a network **20**, and a server **30**. The information processing devices **10**A, **10**B, and **10**C acquire statements of users, detect inputs by the users, and acquire biological information of the users. In addition, the information processing devices **10**A, **10**B, and **10**C have basically similar functions except that arrangement positions are different. Hereinafter, similar descriptions of the information processing devices **10**A, **10**B, and **10**C will be made as a description of an information processing device **10**. The information processing devices **10**A, **10**B, and **10**C are respectively arranged for three users (user A, user B, and user C) participating in a conference. Note that the conference support system **1** only needs to include at least one information processing device although the three information processing devices **10**A, **10**B, and **10**C are arranged in the present embodiment. Furthermore, there may be a plurality of users for one information processing device **10** although a case where there is one user for one information processing device **10** is assumed in the present embodiment. Each of the information processing devices **10** has a function of identifying a plurality of users. The

network **20** connects the information processing devices **10**A, **10**B, and **10**C. The network **20** may be a public communication network, or a communication network not connected to a public wireless communication network. The network **20** may transmit information between the information processing devices **10** by using wireless communication or wired communication. The server **30** is connected to the information processing devices **10**A, **10**B, and **10**C through the network **20**. The server **30** may execute functions of the information processing devices **10**A, **10**B, and **10**C (described later). That is, the server **30** may be connected to the information processing devices **10** through the network and become a part of the information processing devices **10**.
(16) (Information Processing Device)
(17) FIG. **2** is a view illustrating a configuration example of an information processing device according to the present embodiment of the present disclosure. Each of the information processing devices **10** according to the present disclosure includes a user information acquisition unit **100**, an analysis unit **110**, a storage unit **120**, a communication unit **130**, and an output unit **140**. The user information acquisition unit **100** includes a voice information acquisition unit **101**, a biological information acquisition unit **102**, an image information acquisition unit **103**, and a classification information acquisition unit **104**. The analysis unit **110** includes a user information processing unit **111**, a natural language processing unit **112**, a learning unit **113**, an automatic classification unit **114**, a conference progress prediction unit **115**, and a proposal unit **116**. The storage unit **120** includes a user information storage unit **121**, a minutes database unit **122**, a corpus unit **123**, and a dictionary unit **124**. The communication unit **130** includes a transmission unit **131** and a reception unit **132**.
(18) (User Information Acquisition Unit)
(19) The user information acquisition unit **100** includes the voice information acquisition unit **101** that acquires a voice of a user participating in a conference, the biological information acquisition unit **102** that acquires biological information of the user, the image information acquisition unit **103** that acquires an image of a facial expression of the user, and the classification information acquisition unit **104** that acquires, after converting the voice of the user into text data, classification information (annotation label) assigned to the text data by the user. The user information acquisition unit **100** acquires user information of a user participating in a conference while the conference is in progress. Next, components of the user information acquisition unit **100** will be described.
(20) (Voice Information Acquisition Unit)
(21) The voice information acquisition unit **101** acquires a voice of a user participating in a conference. The voice information acquisition unit **101** includes a voice input device such as a microphone installed in a conference room. The voice information acquisition unit **101** outputs, to a voice information processing unit **200**, voice data acquired by the voice input device.
(22) (Biological Information Acquisition Unit)
(23) By using various biological sensors, the biological information acquisition unit **102** acquires biological information of a user participating in the conference. Examples of the biological sensors include a pulse rate meter, a heart rate meter, an electrodermal activity sensor, a sphygmomanometer, a perspiration meter, an activity meter, a blood oxygen level sensor, an infrared sensor, a deep body thermometer, an eye gaze detecting camera, and an eye potential sensor. The biological information acquisition unit **102** acquires data of an index of an autonomic nerve system activity (such as pulse rate, heart rate, skin conductance, blood pressure value, perspiration amount, respiration, or skin temperature/deep body temperature) of the user which index is measured by the biological sensor, and data of an index of an eye gaze activity (such as eye gaze motion/pupil diameter/blink frequency). The biological information acquisition unit **102** only needs to acquire at least one piece of biological information, and the number of pieces of acquired biological information is not limited. The biological information acquisition unit **102** of the present embodiment acquires a pulse rate of the user participating in the conference by using

the pulse rate meter as the biological sensor.
(24) (Image Information Acquisition Unit)
(25) By using a camera, the image information acquisition unit **103** acquires an image of a facial expression of a user participating in the conference. The image information acquisition unit **103** may acquire images of the facial expression of the user from different angles by using a plurality of cameras. The camera includes an optical system, an imaging element such as a complementary metal oxide semiconductor (CMOS) or a charge coupled device (CCD), and an image processing circuit.
(26) (Classification Information Acquisition Unit)
(27) The classification information acquisition unit **104** acquires classification information (annotation label) of text data of a statement which classification information is assigned by a user who checks the text data converted by the analysis unit **110** and output from the output unit **140**. As the classification information assigned to the text data of the statement, there are “problem proposal”, “problem handling”, a “satisfaction level”, and a “problem resolution level”.
(28) The “problem proposal” is the classification information assigned in a case where a user who checks the text data of the statement of the user participating in the conference which text data is output from the output unit **140** determines that the statement proposes a problem to be solved in the conference. The “problem handling” is the classification information assigned in a case where the user who checks the text data of the statement of the user participating in the conference which text data is output to the output unit **140** determines that the statement is a countermeasure to the problem to be solved in the conference. For example, in a case where the classification information of the “problem proposal” is assigned to a statement of a user participating in a conference and another user participating in the conference makes a statement such as “How about contacting Mr. A?” or makes a statement such as “How about checking B?”, the classification information of the “problem handling” is assigned to text data of the statement. Furthermore, the “satisfaction level” included in the classification information is the classification information assigned in a case where the user who checks the text data of the statement of the user participating in the conference which text data is output to the output unit **140** feels satisfied with the statement. The “problem resolution level” included in the classification information is the classification information assigned in a case where it is determined that the problem proposed in the conference is resolved by a statement to which the classification information of the “problem handling” is assigned among statements of the users participating in the conference which statements are output to the output unit **140**. Note that the classification information can be assigned by a user or by automatic classification using a program learned on the basis of determination information of the user, or the like.
(29) The classification information acquisition unit **104** stores the acquired classification information and the text data into the storage unit **120** in association with each other.
(30) (Analysis Unit)
(31) The analysis unit **110** executes analysis on the basis of various kinds of information acquired by the user information acquisition unit **100** and data stored in the storage unit **120**, and executes analysis of a conference held by the conference support system **1** and support of the conference. The analysis unit **110** includes the user information processing unit **111**, the natural language processing unit, the learning unit **113**, the automatic classification unit **114**, the conference progress prediction unit **115**, and the proposal unit **116**. The analysis unit **110** analyzes and learns the user information of the users participating in the conference which user information is acquired by the user information acquisition unit **100**, generates a learned model, and performs inference by using the learned model.
(32) (User Information Processing Unit)
(33) FIG. **3** is a view illustrating a configuration example of a user information processing unit according to the present disclosure. As illustrated in FIG. **4**, the user information processing unit **111** includes a voice information processing unit **200**, a biological information processing unit **300**,

an image information processing unit **400**, and a classification information processing unit **500**.
(34) (Voice Information Processing Unit)
(35) The voice information processing unit **200** performs voice processing on the voice information acquired by the voice information acquisition unit **101**. The voice information processing unit **200** includes a noise elimination unit **202**, a sound source separation unit **204**, an utterance section detection unit **206**, a voice recognition unit **208**, and an acoustic feature amount extraction unit **210**.
(36) The noise elimination unit **202** eliminates noise included in the voice information acquired by the voice information acquisition unit **101**. For example, the noise elimination unit **202** specifies noise from a difference between voice acquired from a microphone arranged near a noise source and voice acquired from other microphones, and eliminates the noise. The noise elimination unit **202** may eliminate noise by using a correlation between voices input to a plurality of microphones.
(37) The sound source separation unit **204** targets voice data from which noise is eliminated by the noise elimination unit **202**, and specifies a sound source (speaker) of each voice. The sound source separation unit **204** specifies a speaker by calculating a direction and distance of each speaker with respect to the microphones from a time difference in which voice is input to the plurality of microphones.
(38) The utterance section detection unit **206** detects a group of utterance by one speaker as an utterance section with respect to the voice data separated by the sound source separation unit **204**. As a method of detecting the utterance section, any known method such as a method using a speech model or spectrum entropy can be used.
(39) The voice recognition unit **208** performs voice recognition processing on a group of voice data detected by the utterance section detection unit **206**, and converts the voice data into text data. Specifically, after being converted into a sound waveform, voice is converted into text data by utilization of a recognition decoder including an acoustic model, a language model, and an utterance dictionary. The acoustic model discriminates voice by using an analysis of a frequency component or a temporal change of a sound waveform. Specifically, a sound waveform is cut out and a feature amount is quantified, and then the discrimination is performed by calculation of to which phoneme model and in which degree the feature amount is close. The language model evaluates whether a character string or a word string verbalized by the acoustic model is appropriate as Japanese. The language model is acquired by collection and statistical processing of a large number of Japanese texts. A pronunciation dictionary is a dictionary that records a relationship between a word and a phoneme string of reading of the word. Since the acoustic model is modeled for each phoneme that is the smallest unit of sound, acoustic models of phonemes are coupled according to the pronunciation dictionary and an acoustic model of a word corresponding to word utterance is configured. That is, after performing an acoustic analysis on the voice data acquired by the voice information acquisition unit **101** by using the recognition decoder in which the acoustic model, the language model, and the utterance dictionary are combined, the voice recognition unit **208** performs conversion into text data by using deep learning. The voice recognition unit **208** outputs the text data that is a result of the voice recognition, and also calculates and outputs reliability of the voice recognition result.
(40) The acoustic feature amount extraction unit **210** calculates an acoustic feature amount (acoustic feature amount) of the voice data. Examples of the acoustic feature amount include a sound pressure level, a fundamental frequency, an utterance length, an utterance speed, an average mora length, overlapping of utterance, an interval of utterance, and the like. In the present embodiment, the acoustic feature amount extraction unit **210** extracts a tone of voice, that is, the fundamental frequency.
(41) With time-series data of the acoustic feature amount calculated by the acoustic feature amount extraction unit **210** being an input, a first satisfaction level calculation unit **212** calculates a satisfaction level of each of users participating in the conference for each unit time on the basis of a

database of a correlation between the acoustic feature amount and the satisfaction level. As data of the correlation between the acoustic feature amount and the satisfaction level, there is a relationship between the fundamental frequency and the satisfaction level. Since there is a correlation in which the satisfaction level becomes high in a case where the fundamental frequency is high, the first satisfaction level calculation unit **212** calculates a first satisfaction level for each of the users participating in the conference from the correlation between the fundamental frequency and the satisfaction level by using the fundamental frequency extracted by the acoustic feature amount extraction unit **210**. For example, the first satisfaction level calculation unit **212** calculates the first satisfaction level as a numerical value in a specific range from 0 (dissatisfaction) to 50 (satisfaction).
(42) (Biological Information Processing Unit)
(43) The biological information processing unit **300** processes the biological information acquired by the biological information acquisition unit **102**. The biological information processing unit **300** includes a second satisfaction level calculation unit **302**.
(44) In the present embodiment, the second satisfaction level calculation unit **302** calculates a second satisfaction level indicating a satisfaction level of each of the users participating in the conference on the basis of a pulse rate acquired by the biological information acquisition unit **102**. Specifically, when the pulse rate increases, it is considered that the user is in a tense state, and the second satisfaction level is calculated to be low. When the pulse rate decreases, it is considered that the user is in a relaxed state, and the second satisfaction level is calculated to be high.
(45) Emotion can be also estimated by a relationship between fluctuation of the pulse rate and the emotion. The emotion may be estimated by recording of data of the relationship between the fluctuation of the pulse rate and the emotion into the storage unit **120**, and collation of fluctuation of the pulse rate acquired by the pulse rate meter with the data of the relationship between the fluctuation of the pulse rate and the emotion which data is recorded in the storage unit **120**.
(46) The second satisfaction level calculation unit **302** calculates the second satisfaction level by the method described above. For example, the second satisfaction level calculation unit **302** may calculate the second satisfaction level as a numerical value in a range from 0 (dissatisfaction) to 50 (satisfaction).
(47) (Image Information Processing Unit)
(48) The image information processing unit **400** processes the image information acquired by the image information acquisition unit **103**. The image information processing unit **400** includes a third satisfaction level calculation unit **402**.
(49) The third satisfaction level calculation unit **402** analyzes emotions of the users participating in the conference by using image information of facial expressions of the users participating in the conference which image information is acquired by the image information acquisition unit **103**. The third satisfaction level calculation unit **402** includes, for example, a facial expression analysis tool such as a facial action coding system (FACS). The facial action coding system (FACS) is a facial expression analysis tool/facial expression theory developed by Paul Ekman, Wallance Friesen, and the like in 1978 to comprehensively measure a visible face movement. In addition, a known analysis technique other than the above technique may be used.
(50) The third satisfaction level calculation unit **402** calculates a third satisfaction level on the basis of a numerical value of an emotion of “joy” classified as a positive emotion and numerical values of emotions of “disgust”, “anger”, “sadness”, “contempt”, “fear”, and “surprise” classified as negative emotions among the calculated emotions. The third satisfaction level calculation unit **402** may calculate the third satisfaction level as a value in a range from 0 (dissatisfaction) to 50 (satisfaction).
(51) (Classification Information Processing Unit)
(52) The classification information processing unit **500** processes the classification information acquired by the classification information acquisition unit **104**. The classification information

processing unit **500** includes a problem resolution level calculation unit **502** and a fourth satisfaction level calculation unit **504**.

(53) The problem resolution level calculation unit **502** calculates a problem resolution level indicating a degree to which a problem proposed in the conference is resolved by a statement of a user participating in the conference (statement to which classification information of problem handling is assigned). The problem resolution level calculation unit **502** counts the number of users who assign the classification information of the problem resolution level to the text data to which the classification information of the problem handling is assigned. In a case where all the users participating in the conference assign the classification information of the problem resolution level, the problem resolution level is calculated as 100. In a case where a part of the users participating in the conference assigns the classification information of the problem resolution level, a value acquired by division of the number of users who assign the classification information of the problem resolution level by the number of all the users participating in the conference is multiplied by 100, whereby the problem resolution level is calculated. In a case where none of the users participating in the conference assigns the classification information of the problem resolution level, the problem resolution level calculation unit **502** calculates the problem resolution level as 0. As a result, the problem resolution level calculation unit **502** calculates the problem resolution level with respect to the statement to which the problem handling is assigned.

(54) The fourth satisfaction level calculation unit **504** counts, with respect to each of the users participating in the conference, the number of times of assignment of the classification information of the satisfaction level. The fourth satisfaction level calculation unit **504** calculates the fourth satisfaction level of a user who participates in the conference and assigns the classification information of the satisfaction level as 50 (satisfaction), and calculates the fourth satisfaction level of a user who does not assign the classification information of the satisfaction level as 0 (dissatisfaction).

(55) (Satisfaction Level Calculation Unit)

(56) A satisfaction level calculation unit **600** calculates a total value of the satisfaction levels of each of the users participating in the conference by adding up, for each of the users participating in the conference, the first satisfaction level calculated by the first satisfaction level calculation unit **212** included in the voice information processing unit **200**, the second satisfaction level calculated by the second satisfaction level calculation unit **302** included in the biological information processing unit **300**, the third satisfaction level calculated by the third satisfaction level calculation unit **402** included in the image information processing unit **400**, and the fourth satisfaction level calculated by the fourth satisfaction level calculation unit **504** included in the classification information processing unit **500**.

(57) (Natural Language Processing Unit)

(58) The natural language processing unit **112** performs natural language processing including a morphological analysis, syntax analysis, context analysis, and knowledge acquisition on the text data acquired by conversion of the voice information of the users participating in the conference by the voice information processing unit **200**. In the present embodiment, with a statement at the conference being an input, the natural language processing unit **112** determines whether a statement of an opinion related to a subject is made.

(59) FIG. **4** is a view illustrating a configuration example of the natural language processing unit according to the present disclosure. As illustrated in FIG. **5**, the natural language processing unit **112** includes a morphological analysis unit **1121**, a syntax analysis unit **1122**, a context analysis unit **1123**, and a knowledge acquisition unit **1124**.

(60) The morphological analysis unit **1121** executes the morphological analysis on a text included in the text data acquired by conversion of the voice information. The morphological analysis is an analysis of dividing a sentence included in the text data into a string of morphemes on the basis of a grammar of a target language and information such as a part of speech of a word, which

information is called a dictionary, from the text data in a natural language without a note of grammatical information, and discriminating a part of speech or the like of each morpheme. For example, MeCab can be used as the morphological analysis unit **1121**. In addition, a morphological analysis engine JUMAN may be used.

(61) The syntax analysis unit **1122** performs the syntax analysis on the data divided into the string of morphemes by the morphological analysis unit **1121**. The syntax analysis unit **1122** executes the syntax analysis by using information stored in the corpus unit **123** of the storage unit **120**. The syntax analysis is a method of analyzing a structure of a sentence from text data in a natural language. That is, the syntax analysis unit **1122** specifies a structure of a sentence by organizing a positional relationship (such as relationship of dependency) between morphemes included in a sentence of the text data in the natural language, and expressing the positional relationship with a syntax tree. As the syntax analysis unit **1122**, for example, Cabocha, KNP, or the like using a support vector machine is used. Cabocha is syntax analysis software used in combination with Mecab that is a morphological analysis engine. With an analysis result (morpheme string) of JUMAN that is a morphological analysis engine being an input, KNP analyzes a dependency relationship, a case relationship, and an anaphoric relationship between clauses and basic phrases.

(62) With respect to the text data acquired by conversion of the voice information, the context analysis unit **1123** executes the context analysis on a statement to which the classification information of the “problem handling” corresponding to the statement labeled with the classification information of the “problem proposal” by the users participating in the conference is assigned. That is, the context analysis unit **1123** analyzes a relationship between a sentence of the statement to which the classification information of the “problem proposal” is assigned and a sentence of the statement to which the classification information of the “problem handling” is assigned. Bidirectional encoder representations from transformers (BERT) can be used as the context analysis unit **1123**. In addition, XLNet having a property of simultaneously handling forward and backward information while using an autoregressive language model that learns dependency between words to be predicted, A Light BERT (ALBERT) that is a lightweight version of BERT, or the like may be used.

(63) The knowledge acquisition unit **1124** executes the knowledge acquisition from a text included in data acquired by conversion of the voice information acquired by the voice information acquisition unit **101** into the text data. The knowledge acquisition is a natural language processing technique of automatically acquiring knowledge from texts of text data in a natural language. The knowledge acquisition unit **1124** can acquire knowledge from text data included in minutes data **122** and store the acquired knowledge as information corresponding to the text data of the minutes data **122**.

(64) (Learning Unit)

(65) The learning unit **113** includes a first learning unit **1131** and a second learning unit **1132**. The learning unit **113** creates training data created from data of user information acquired from the users participating in the conference and minutes data generated by utilization of the user information, and generates a learned model. The learning unit **113** may create a learned model by learning, as unsupervised data, data of user information acquired from the users participating in the conference and minutes data generated by utilization of the user information. Each of the first learning unit **1131** and the second learning unit **1132** creates a learned model. The number of learned models created by each of the first learning unit **1131** and the second learning unit **1132** is not limited to one, and a plurality of learned models may be included on the basis of specifications thereof, a purpose, information to be input, and information to be output.

(66) The first learning unit **1131** learns relevance between the text data acquired by conversion of the voice information, which is acquired by the user information acquisition unit **100**, by the voice information processing unit **200** and the classification information assigned by the users, and creates a first learned model. The first learning unit **1131** processes text data of statements of the

users during the conference by the first learned model, and determines classification information with respect to the text data. That is, the first learning unit **1131** creates a learned model in which a characteristic how a user participating in the conference assigns classification information (problem proposal, problem handling, satisfaction level, and problem resolution level) to text data of a statement of a user (another user) participating in the conference, and assigns the classification information with respect to the statement by using the learned model. BERT can be used as the first learning unit **1131**. BERT is a distributed representation computation mechanism that can be used generally with respect to various kinds of natural language processing. Note that the first learning unit **1131** of the present embodiment performs supervised learning with the training data to which the classification information is assigned by the users. However, a learned model may be created by unsupervised learning with provision of the classification information. In this case, after data is accumulated in the minutes database unit **122** of the storage unit **120**, relearning may be performed with the accumulated data as training data.

(67) The second learning unit **1132** learns minutes data **1221** recorded in the minutes database unit **122** of the storage unit **120** by using unsupervised learning (such as K-MEANS clustering, for example), and generates a second learned model. The unsupervised learning is a method of generating a learned model by extracting a relationship between pieces of data without a label by using predetermined data. The second learning unit **1132** learns a feature amount of the minutes data **1221** and generates the second learned model of a relationship between pieces of data included in the minutes data **1221**. The second learned model learns a status of the conference on the basis of a statement, an evaluation input with respect to the statement, and information of various reactions of the users, the statement, evaluation, and information being included in the minutes data **1221**. In the second learned model, user information of the currently ongoing conference which user information is acquired by the user information acquisition unit **100** is an input, and an evaluation of the conference is an output.

(68) With the user information that includes the voice information, the biological information, the image information, and the classification information and that is acquired by the user information acquisition unit **100** being an input, the second learned model infers the data included in the minutes data **1221**. For example, when the voice information processing unit **200** extracts the fundamental frequency from the voice information acquired by the voice information acquisition unit **101** included in the user information acquisition unit **100** and inputs the fundamental frequency to the second learned model, data of relevance information **1227** included in the minutes data **1221** is inferred from a feature amount of the fundamental frequency. In addition, accuracy of the inference can be made higher when an amount of information input to the second learned model is larger. That is, when the biological information, the image information, and the classification information are further input to the second learned model generated by the second learning unit **1132** in addition to the voice information acquired by the user information acquisition unit **100**, it is possible to increase the accuracy of the inference of the relevance information **1227**.

(69) More specifically, in the second learned model, when the voice information of the users participating in the conference (including information converted into text data by the voice information processing unit **200**), the pulse rates of the users, and the image information of the facial expressions of the users are input, the relevance information **1227** assigned to the currently ongoing conference is calculated. That is, the second learned model calculates to which one of a “successful conference” and an “unsuccessful conference” and to a conference classified into which pattern the currently ongoing conference corresponds. Thus, by using the second learned model, it is possible to determine whether the currently ongoing conference is likely to be the “successful conference” or the “unsuccessful conference” on the basis of the user information of the currently ongoing conference.

(70) (Automatic Classification Unit)

(71) The automatic classification unit **114** automatically classifies text data of statements of the

users of the currently ongoing conference by using the first learned model created by the first learning unit **1131**. The automatic classification unit **114** inputs the text data of the statements of the users of the currently ongoing conference into the first learned model of the first learning unit **1131**, and acquires classification information output from the first learning unit **1131**. As a result, the automatic classification unit **114** acquires information indicating which one of the “problem proposal”, the “problem handling”, and “others” the text data is classified into.
(72) Furthermore, in a case where the text data of the statement is classified into the “problem handling”, by using the first learned model, the automatic classification unit **114** calculates, with respect to the text data of the “problem handling”, how many users among the users participating in the conference assign the classification information of the “satisfaction level” to the text data. The automatic classification unit **114** transmits an inferred result to the output unit **140**. In such a manner, the automatic classification unit **114** automatically classifies the statements of the users at the currently ongoing conference.
(73) (Conference Progress Prediction Unit)
(74) The conference progress prediction unit **115** predicts progress of the currently ongoing conference (whether the conference is likely to be the “successful conference” or the “unsuccessful conference”) by using the second learned model of the second learning unit **1132**. The conference progress prediction unit **115** inputs the user information acquired by the user information acquisition unit **100** into the second learning unit **1132**, and acquires a determination result that is calculated by the second learned model and that indicates whether the currently ongoing conference is likely to be the “successful conference” or the “unsuccessful conference”. The conference progress prediction unit **115** calculates a problem resolution level prediction value on the basis of information on the determination result.
(75) (Proposal Unit)
(76) By using the second learned model of the second learning unit **1132**, the proposal unit **116** proposes “problem handling” to a “problem proposal”-assigned statement, which is made by a user participating in the currently ongoing conference, with reference to minutes data **1221** of a similar conference which data is included in the minutes database unit **122**. In addition, in a case where the conference progress prediction unit **115** predicts that the currently ongoing conference becomes the “unsuccessful conference”, the proposal unit **116** proposes, by using the second learned model of the second learning unit **1132**, a “topic change”, a “problem proposal” assumed in the future, “problem handling” with respect to a “problem proposal” in the currently ongoing conference, and the like to the users participating in the conference.
(77) (Storage Unit)
(78) The storage unit **120** may use a temporary storage medium such as a read only memoly (ROM) or a ramdom access memoly (RAM). In addition, a magnetic storage medium such as a magnetic tape or a hard disk drive (HDD) may be used, and a nonvolatile memory such as an ellectrically erasable programmable read only memoly (EEPROM), a flash memoly, a magnetoresistive ramdom access memoly (MRAM), a ferroelectric ramdom access memoly (FeRAM), or a phase change ramdom access memoly (PRAM) may be used.
(79) The storage unit **120** includes the user information storage unit **121**, the minutes database unit **122**, the corpus unit **123**, and the dictionary unit **124**. The user information storage unit **121** records the user information acquired by the user information acquisition unit **100** including the voice information acquisition unit **101**, the biological information acquisition unit **102**, and the image information acquisition unit **103**.
(80) (Minutes Database Unit)
(81) The minutes database unit **122** stores minutes data **1221** of a past conference. The data included in the minutes data **1221** may be only text data of statements of the users participating in the conference, or may be information in which the classification information of the “problem proposal”, the “problem handling”, the “satisfaction level”, and “problem resolution” is assigned to

the text data.

(82) FIG. **5** is a schematic diagram illustrating information included in the minutes data. The minutes data **1221** illustrated in FIG. **5** is data including conference information **1222**, voice information **1223**, biological information **1224**, image information **1225**, classification information **1226**, and relevance information **1227**. The conference information **1222** is data including a conference date and time, a conference place, conference participants, a subject, and the like. The voice information **1223** is data generated when the voice information acquisition unit **101** acquires voices of the users participating in the conference. The biological information **1224** is data generated when the biological information acquisition unit **102** acquires biological information of the users participating in the conference. The image information **1225** is data generated when the image information acquisition unit **103** acquires image information of facial expressions of the users participating in the conference. The classification information **1226** is data generated when the classification information acquisition unit **104** acquires the classification information assigned by the users participating in the conference. Note that the biological information **1224** is a tone of voice, that is, the fundamental frequency, and the biological information **1224** is a pulse rate in the present embodiment.

(83) The relevance information **1227** includes information indicating whether the conference is successful or unsuccessful. The information indicating whether the conference is successful or unsuccessful may be classified according to reasons why the conference is unsuccessful. As patterns in which the conference becomes unsuccessful, for example, there are a case where the same “subject” is set again and a conference is held since users participating in the conference make no statement, a case where the same “subject” is set again and a conference is held since users participating in the conference repeat statements in a different field with respect to a “subject” and “problem proposal” and discussion is not settled, a case where a part of participants is not actually satisfied and a discussion on the same “subject” is repeated in another conference although all the participants in the conference apparently agree with “problem handling” proposed in the conference, a case where a conference is held on the same “problem” again since the “problem” is not solved although all participants of the conference agree with “problem handling” stated by a user in the conference and the “problem handling” that is regarded as a conclusion of the conference is executed, and a case where users participating in a conference cannot notice a “problem” to be solved in the conference and notices the “problem” to be solved in the conference later and a conference is set for the “problem”.

(84) Furthermore, the relevance information **1227** also includes information related to relevance between minutes data of an individual conference and minutes data of another conference. The relevance is information such as conferences with the same subject, and conferences with the same participants. In addition, with respect to each of the above-described unsuccessful conferences and conferences held again because of those, an unsuccessful pattern and the relevance may be associated. The information that is related to the relevance of individual conference and that can be assigned as the relevance information is not limited to the above, and the users may provide arbitrary relevance information.

(85) (Corpus Unit)

(86) The corpus unit **123** is acquired by structuring of texts in a natural language into language data that can be searched by a computer and integration of the language data on a large scale. For example, a Japanese corpus called KOTONOHA developed by the National Institute for Japanese Language and Linguistics can be used. The data stored in the corpus unit **123** is used in the processing by the natural language processing unit **112**.

(87) (Dictionary Unit)

(88) The dictionary unit **124** is a database in which knowledge necessary for natural language processing regarding words is accumulated. The dictionary unit **124** may record meaning, pronunciation, right and left connectivity, inflection, notation fluctuation, surface case frame, and

the like of the words. As the dictionary unit **124**, for example, an IPA dictionary (ipadic) mounted on Mecab as a standard function can be used. The data stored in the dictionary unit **124** is used in the processing by the natural language processing unit **112**.

(89) (Communication Unit)

(90) The communication unit **130** includes a transmission unit **131** and a reception unit **132**. The communication unit **130** performs data communication by wireless communication or wired communication. In a case of the wireless communication, a communication antenna, an RF circuit, and another communication processing circuit may be included. In a case of the wired communication, for example, a LAN terminal, a transmission circuit, and another communication processing circuit may be included. The information processing device **10** is connected to the network **20** through the communication unit **130**.

(91) (Output Unit)

(92) The output unit **140** visually or auditorially notifies the users of information related to the conference. As the output unit **140**, a cathode-ray tube (CRT) display device, a liquid crystal display device, a plasma display device, an organic electro luminescence (EL) display device, a micro light emitting diode (LED) display device, or the like can be used as a device that gives visual notification. A multifunction machine having a printing function may also be used. That is, the conference may be supported by printing and outputting of information related to the conference on paper. In a case where notification is given auditorially, the output unit **140** may be a speaker, a headphone, or an earphone. The earphone may be a wireless earphone. In a case of the wireless earphone, information related to the conference may be received from the information processing device **10** by utilization of a communication means such as BLUETHOOTH (registered trademark), and the information related to the conference may be output as sound.

(93) Next, processing by the conference support system **1** and the information processing device **10** will be described in the following order. 1. Acquisition of user information at the time of progress of a conference 2. Analysis and learning based on user information 3. Analysis of user information, and creation of classification information based on learning 4. Analysis of user information, and evaluation and proposal based on learning

1. Acquisition of User Information According to Progress of a Conference

(94) The conference support system **1** acquires user information at the time of a conference. Hereinafter, processing in which the conference support system **1** acquires user data from an ongoing conference and further assigns classification information on the basis of an input by a user will be described. The conference support system **1** also acquires user data in a case of determining support to be provided to the ongoing conference.

(95) In the information processing device **10**, the voice information acquisition unit **101** acquires voice, and the voice information processing unit **200** of the user information processing unit **111** performs processing, specifies a user who makes a statement, and generates text data from the voice. The information processing device **10** outputs the created text data to the output unit **140** in association with the user. The information processing device **10** detects classification information input with respect to an image displayed on the output unit **140**.

(96) FIG. **6** is a schematic diagram of a conference progress status display displayed on the output unit of the information processing device. An example of classification information assignment processing will be described with reference to FIG. **6**. When a first user participating in a conference states “Is there any method for solving a problem A? What shall we do?”, the voice information acquisition unit **101** acquires voice information of the statement. The voice information processing unit **200** converts the voice information into text data, and outputs text data **50**A**1** of the statement to the output unit **140** together with a mark **40**A of the first user.

(97) Then, when a second user who listens to the text data **50**A**1** states “How about using a method B?”, the voice information acquisition unit **101** acquires voice information of the statement of the second user, and the voice information processing unit **200** converts the voice information into text

data and outputs text data **50**B**1** of the statement to the output unit **140** together with a mark **40**B of the second user.
(98) Then, when the first user states "I see. That is a good idea", the voice information acquisition unit **101** acquires voice information of the statement of the first user, and the voice information processing unit **200** converts the voice information into text data **50**C**1** and outputs statement text data **50**A**2** to the output unit **140** together with the mark **40**A.
(99) Then, when a third user states "I see. That is a good idea", the voice information acquisition unit **101** acquires voice information of a statement of a user C, and the voice information processing unit **200** converts the voice information into text data, and outputs text data **50**C**1** of the statement to the output unit **140** together with a mark **40**C of the third user. As described above, the information processing device **10** detects a statement of each user, performs a display thereof as text data, and makes it possible to check the statements in time series.
(100) When checking the text data **50**A**1** output to the output unit **140** and determining that the text data **50**A**1** corresponds to a "problem proposal", users participating in the conference input classification information indicating that the text data **50**A**1** corresponds to the "problem proposal". When acquiring the classification information indicating the correspondence to the "problem proposal", the classification information acquisition unit **104** assigns a mark **70**A indicating the "problem proposal" to the text data **50**A**1** displayed on the output unit **140**, and causes the output unit **140** to perform a display thereof.
(101) In addition, in a case of checking the text data **50**B**1** of a statement of a user B which text data is output to the output unit **140**, and determining that the text data **50**B**1** corresponds to "problem handling", the users participating in the conference input classification information of the "problem handling". When acquiring the classification information of the "problem handling", the classification information acquisition unit **104** assigns a mark **70**B indicating the "problem handling" to the text data **50**B**1** output to the output unit **140**, and causes the output unit **140** to perform a display thereof.
(102) In addition, in a case where the mark **70**B indicating the "problem handling" is output to the output unit **140**, the users participating in the conference can assign classification information of a "satisfaction level" and a "problem resolution level" to the text data **50**B**1** of the statement of the user B to which text data the mark **70**B of the "problem handling" is assigned. In a case of acquiring the classification information of the "satisfaction level", the classification information acquisition unit **104** assigns a mark **70**C indicating the "satisfaction level" to the text data **50**B**1** output to the output unit **140**, and causes a display thereof. In a case of acquiring the classification information of the "problem resolution level", the classification information acquisition unit **104** assigns a mark **70**D indicating the "problem resolution level" to the text data **50**B**1** output to the output unit **140**, and causes the output unit **140** to perform a display thereof.
(103) In addition, in the information processing device **10**, the user information processing unit **111** calculates a satisfaction level on the basis of voice information, biological information, and image information of the users participating in the conference, and displays a graph **41** of the calculated satisfaction level. The graph **41** indicates a temporal change in the satisfaction level with a horizontal axis representing time and a vertical axis representing the satisfaction level. The satisfaction level calculation unit **600** calculates an "entire conference satisfaction level" by adding up, with respect to all the users participating in the conference, total values of the satisfaction levels which total values are respectively calculated for the users participating in the conference. As a result, it becomes possible to grasp the satisfaction levels of the users participating in the currently ongoing conference while the conference is in progress, and it is possible to support management of the conference.
(104) In such a manner, the information processing device **10** can make it possible to easily identify contents of the conference by setting the problem proposal, the problem handling, the satisfaction level, and the problem resolution level as the classification information and making it possible to

assign the classification information to text data on the basis of inputs by the users. Furthermore, assignment of the classification information is not necessarily performed during the conference, and an input thereof may be performed after the conference. The information processing device **10** stores the acquired text information, classification information, and the like in the minutes database unit **122**.

(105) 2. Analysis and Learning Based on User Information

(106) The information processing device **10** performs learning in the first learning unit **1131** by using the information in the minutes database unit **122**, and creates a first learned model. Specifically, learning is performed by utilization of text data generated from voice in the minutes database unit **122** and classification information assigned to the text data, and a first learning model in which the text data is an input and the classification information is an output is created. Furthermore, the first learning unit **1131** may use biological information and image information as inputs in addition to the text data. As a result, the first learning model classifies whether a statement of a user is a problem proposal or a solution proposal. In addition, the first learned model also assigns a satisfaction level with respect to the statement of the user. By adjusting a data set at the time of the learning, the first learning unit **1131** can set information to be assigned as the classification information depending on a use.

(107) The information processing device **10** performs learning in the second learning unit **1132** by using the information in the minutes database unit **122**, and creates a second learned model. The second learning unit **1132** performs learning with the user information including the voice information, the biological information, the image information, and the classification information acquired by the user information acquisition unit **100** being an input and with the minutes data **1221** being an output, and creates a second learning model. An input of the second learned model includes at least the voice information and the classification information, and preferably includes the biological information and the image information furthermore. In addition, relevance information to be output is whether an ongoing conference is the "successful conference" or the "unsuccessful conference", and to which of patterns the conference corresponds in a case of the unsuccessful conference. Furthermore, the relevance information also includes information of a conference similar to the ongoing conference.

(108) 3. Analysis of User Information, and Creation of Classification Information Based on Learning

(109) The information processing device **10** creates classification information with respect to the ongoing conference by using the automatic classification unit **114** that executes processing by the first learned model. The automatic classification unit **114** acquires information indicating which one of the "problem proposal", the "problem handling", and "others" the text data is classified into. Furthermore, in a case where the text data of the statement is classified into the "problem handling", by using the first learned model, the automatic classification unit **114** calculates, with respect to the text data of the "problem handling", how many users among the users participating in the conference assign the classification information of the "satisfaction level" to the text data. By executing the above processing, the classification information assigned by the users in the above-described processing of FIG. **6** can be assigned by the automatic classification unit **114**. By classifying the statement of the users, the automatic classification unit **114** can notice a statement of a "problem" or "problem handling" overlooked by the users participating in the conference and can support management of the conference. In addition, the automatic classification unit **114** can calculate a satisfaction level of the "problem handling" by using the first learning model, and can provide more useful information as reference information of the conference.

(110) The information processing device **10** can process the text data (statement) classified by the automatic classification unit **114** by the natural language processing unit **112**, evaluate contents of the statement, and output a result thereof.

(111) FIG. **7** is a flowchart for describing a procedure of information processing by the natural

language processing unit. Processing in which the natural language processing unit **112** evaluates contents of the text classified as the problem proposal by the automatic classification unit **114** is illustrated in FIG. **7**. The context analysis unit **1123** of the natural language processing unit **112** performs a semantic analysis of meaning of text data of a "problem proposal"-assigned statement by using meaning of words recorded in the dictionary unit **124** (Step S**202**). On the basis of a result of the semantic analysis, the context analysis unit **1123** specifies a "range of problem handling" required by the "problem proposal" (Step S**204**). Similarly, the context analysis unit **1123** analyzes meaning of text data of a "problem handling"-assigned statement (Step S**206**). It is determined whether the meaning of the text data of the "problem handling"-assigned statement is included in the "range of problem handling" described above (Step S**208**). In a case where the meaning of the text data of the "problem handling"-assigned statement is included in the "range of problem handling" described above (Step S**208**; Yes), the context analysis unit **1123** determines that the context is positive (Step S**210**). In a case where the meaning of the text data of the "problem handling"-assigned statement is not included in the "range of problem handling" (Step S**208**; No), the context analysis unit **1123** determines that the context is negative (Step S**212**).
(112) On the basis of information of results of the analysis, the information processing device **10** can evaluate a statement and acquire an evaluation result. By outputting a determination result indicating whether an appropriate "problem handling"-assigned statement is made with respect to a "problem proposal"-assigned statement of a user participating in the currently ongoing conference, it becomes possible to check an evaluation of the statement in the conference. Thus, it is possible to support management of the conference.
(113) 4. Analysis of User Information, and Evaluation and Proposal Based on Learning
(114) The information processing device **10** acquires the user information of the currently ongoing conference, and the conference progress prediction unit **115** evaluates whether the conference becomes successful or not by processing the acquired information by the second learned model. FIG. **8** is a description diagram for describing definitions of the "successful conference" and the "unsuccessful conference". As illustrated in FIG. **8**, the "unsuccessful conference" is classified into a pattern A to pattern E, and the "successful conference" is defined as a pattern F and a pattern G in the present embodiment.
(115) The conference of the pattern A is a case where the users participating in the conference give no opinion on a "subject". That is, this is a case where the users participating in the conference make no statement.
(116) The conference of the pattern B is a case where the users participating in the conference repeat statements in a different field with respect to a "subject" of the conference and discussion is not settled.
(117) The conference of the pattern C is a case where a user participating in the conference makes, with respect to a "problem proposal", a statement classified as "problem handling" included in a "range of problem handling" required by the "problem proposal" and discussion is coming into conclusion to perform the "problem handling", but actually, the problem proposed by the "problem proposal"-assigned statement cannot be solved even when the "problem handling" is performed. In other words, it can be said that this is a case where it is found that the problem proposed by the "problem proposal"-assigned statement cannot be solved when the "problem handling" that is the conclusion of the conference is actually performed after the conference.
(118) The conference of the pattern D is defined as a case where a part of the users participating in the conference is not actually satisfied although all the users participating in the conference apparently agree with "problem handling" proposed in the conference, and a discussion on the same "subject" is repeated in another conference. The conference of the pattern E is defined as a case where the users participating in the conference cannot notice a "problem" to be solved in the conference and later notice that the "problem" is to be solved in the conference, and a conference is set for the "problem". A case where a conference applies to the definitions of the conferences of the

pattern A to the pattern E described above is defined as the "unsuccessful conference".
(119) The conference of the pattern F is defined as a case that does not apply to the definitions of the conferences of the pattern A to the pattern E and a pattern G. The conference of the pattern G is a case where the second satisfaction level calculated by the second satisfaction level calculation unit **302** included in the biological information processing unit **300** with respect to the biological information acquired by the biological information acquisition unit **102** is 25 or higher (positive), the context analysis unit **1123** determines that a relationship of a "problem handling"-assigned statement corresponding to a "problem proposal"-assigned statement is positive after the voice information processing unit **200** converts the voice information acquired by the voice information acquisition unit **101** into the text data, and the number of the users who participate in the conference and assign the classification information of the "satisfaction level" and the "problem resolution level" to the "problem handling"-assigned statement is equal to or larger than a majority of the users participating in the conference.
(120) FIG. **9** is a flowchart illustrating a procedure of information processing by the conference progress prediction unit according to the present disclosure. A procedure of the information processing by the conference progress prediction unit **115** will be described with reference to FIG. **9**. The user information acquired by the user information acquisition unit **100** with respect to the currently ongoing conference is input to the second learned model included in the conference progress prediction unit **115**, and relevance information of the currently ongoing conference is inferred (Step S**102**). On the basis of the relevance information output by the second learned model, it is determined whether the currently ongoing conference corresponds to the "unsuccessful conference" (Step S**104**).
(121) In a case where the conference progress prediction unit **115** determines that the currently ongoing conference corresponds to the "unsuccessful conference" (Step S**104**; YES), it is checked to which pattern the relevance information inferred by the second learned model applies (Step S**106**). In a case where the relevance information is the pattern A (Step S**106**; pattern A), the conference progress prediction unit **115** calculates a problem resolution level prediction value as 0 (Step S**110**). In a case where the relevance information is the pattern B (Step S**106**; pattern B), the conference progress prediction unit **115** calculates a problem resolution level prediction value as 10 (Step S**112**). In a case where the relevance information is the pattern C (Step S**106**; pattern C), the conference progress prediction unit **115** calculates a problem resolution level prediction value as 30 (Step S**114**). In a case where the relevance information is the pattern D (Step S**106**; pattern D), the conference progress prediction unit **115** calculates a problem resolution level prediction value as 50 (Step S**116**). In a case where the relevance information is the pattern E (Step S**106**; pattern E), the conference progress prediction unit **115** calculates a problem resolution level prediction value as 50 (Step S**118**).
(122) In a case of determining that the currently ongoing conference does not correspond to the "unsuccessful conference" (Step S**104**; NO), the conference progress prediction unit **115** determines whether the biological information acquired by the biological information acquisition unit **102** from the users in the currently ongoing conference is positive (that is, second satisfaction level calculated by the second satisfaction level calculation unit **302** included in the biological information processing unit **300** is 25 or higher (positive)), the context analysis unit **1123** determines a relationship of the "problem handling"-assigned statement corresponding to the "problem proposal"-assigned statement is positive, and the number of users who assign the "satisfaction level" and the "problem resolution level" is equal to or larger than the majority of the users participating in the conference (Step S**108**). In a case where Step S**108** is YES (Step S**108**; Yes), the conference progress prediction unit **115** calculates a problem resolution level as 100 (Step S**120**).
(123) In a case where Step S**108** is No (Step S**108**; No), the conference progress prediction unit **115** calculates a problem resolution level as 60 (Step S**122**).

(124) As described above, the conference progress prediction unit **115** calculates the problem resolution level prediction value according to the relevance information inferred by the second learned model. By outputting the problem resolution level prediction value, the information processing device **10** can notify the users of the evaluation result of the problem resolution level of the proposal in the currently ongoing conference, and can support the progress of the conference.
(125) The information processing device **10** performs classification on the basis of the user information of the currently ongoing conference, and the proposal unit **116** makes a proposal on the basis of a classification result. FIG. **10** is a flowchart illustrating a procedure of information processing by the proposal unit. The information processing by the proposal unit **116** will be described in detail with reference to FIG. **10**. When the classification information of the "problem proposal" is assigned to text data of a statement, the proposal unit **116** searches text data to which the classification information of the "problem proposal" is assigned and which is included in past minutes data **1221** recorded in the minutes database unit **122**, and checks whether there is minutes data **1221** including text data of a past "problem proposal"-assigned statement similar to the "problem proposal"-assigned statement in the currently ongoing conference (Step S**302**). In a case where there is the minutes data **1221** including the similar "problem proposal"-assigned statement in the minutes database unit **122** (Step S**302**; Yes), the proposal unit **116** checks whether the minutes data **1221** including the "problem proposal"-assigned statement includes text data of a "problem handling"-assigned statement corresponding to the "problem proposal"-assigned statement (Step S**304**). In a case where there is the text data of the "problem handling"-assigned statement corresponding to the "problem proposal"-assigned statement (Step S**304**; Yes), the proposal unit **116** extracts the text data of the "problem handling"-assigned statement (Step S**306**). The proposal unit **116** counts the number of times of assignment of the classification information of the "satisfaction level" and the "problem resolution level" to the extracted text data of the "problem handling"-assigned statement (Step S**308**). The proposal unit **116** ranks pieces of the extracted text data of the "problem handling"-assigned statements in descending order of the number of times of assignment of the classification information of the "satisfaction level" and the "problem resolution level" (Step S**310**). The proposal unit **116** outputs, to the output unit **140**, the top three pieces of the text data of the "problem handling"-assigned statements in the rank (Step S**312**).
(126) In a case where the minutes data **1221** including the "problem proposal"-assigned statement similar to the "problem proposal"-assigned statement in the currently ongoing conference does not exist in the minutes database unit **122** (Step S**302**; No), the proposal unit **116** ends the information processing. In a case where the text data of the "problem handling"-assigned statement corresponding to the "problem proposal"-assigned statement does not exist in the minutes data **1221** extracted in Step S**304** (Step S**304**; No), the proposal unit **116** ends the information processing.
(127) The information processing device **10** can support resolution of the "problem" in the currently ongoing conference by selecting a proposal on the basis of contents of the classified problem and information of the minutes data in the past by using the first learning model. Furthermore, the information processing device **10** may execute learning in the proposal unit **116** and create a proposal acquired by arrangement of the past proposal on the basis of past information.
(128) Here, the determination criterion of the patterns is an example, and is not limited to the above. Regarding whether the currently ongoing conference is the conference classified into the pattern A, the currently ongoing conference may be predicted as the conference classified into the pattern A in a case where a determination result of the context analysis unit **1123** of the natural language processing unit **112** is negative. Regarding whether the currently ongoing conference is a conference classified into the pattern B, the currently ongoing conference may be predicted as the conference classified into the pattern C in a case where the voice information processing unit **200** converts voice information acquired from the users participating in the conference into text data and text data of a statement is not generated for 10 minutes or more. In addition, regarding whether

the currently ongoing conference is a conference classified into the pattern D, the currently ongoing conference may be predicted as the conference classified into the pattern D in a case where there is at least one user whose second satisfaction level and fourth satisfaction level are not consistent among the users participating in the conference.

(129) FIG. **11** is a description diagram for describing a function of the proposal unit. The function of the proposal unit **116** will be described with reference to FIG. **11**. The first user participating in the conference states "Is there any method for solving a problem A? What shall we do?", and a mark **70**A of the classification information of the "problem proposal" is assigned to text data **50**A**3** of the statement together with the mark **40**A of the first user.

(130) In the conference support system **1**, the analysis unit **110** analyzes the user information, and the satisfaction level calculation unit **600** calculates the entire conference satisfaction level. In a case where the calculated entire conference satisfaction level is decreased, the proposal unit **116** infers a proposal for a conference body.

(131) The proposal unit **116** executes processing on the text data **50**A**3** and creates the proposal. The proposal unit **116** outputs text data **50**E**1** "There are XX similar problem handling cases in the past" together with a mark **43**. Furthermore, the proposal unit **116** outputs proposals **60**A and **60**B corresponding to the text data **50**A**3** of the "problem proposal". Here, the proposal unit **116** displays the proposals **60**A and **60**B in order of ranks.

(132) In a case of determining that the proposal **60**A and the proposal **60**B can resolve the problem, the users participating in the conference assign the classification information of the "problem resolution level" thereto. In addition, in a case where the users participating in the conference feel satisfied with the "problem handling", the classification information of the "satisfaction level" is assigned. In the present embodiment, since the classification information of the "problem resolution level" and the "satisfaction level" is assigned to the proposal **60**A, a mark **70**C indicating the "problem resolution level" and a mark **70**D indicating the "satisfaction level" are displayed below the proposal **60**A.

(133) In such a manner, the information processing device **10** can smoothly progress the conference by making a proposal for the problem. In addition, the information processing device **10** stores, into the minutes database unit **122**, the classification information input with respect to the proposal during the conference. As a result, usage for learning in the first learning unit **1131** is possible.

(134) Furthermore, the information processing device **10** may evaluate the currently ongoing conference on the basis of user information of the conference, and make a proposal by the proposal unit **116** in a case of the unsuccessful conference. FIG. **12** is a flowchart illustrating a procedure of information processing by the proposal unit **116**. A procedure of another example of the information processing by the proposal unit **116** will be described with reference to FIG. **12**. The information processing device **10** determines, by a problem resolution level prediction unit **1152**, whether the currently ongoing conference is the "successful conference" (Step S**402**).

(135) In the information processing device **10**, in a case where the conference progress prediction unit **115** predicts that the currently ongoing conference corresponds to the "successful conference" (Step S**404**; Yes), the proposal unit **116** ends the information processing.

(136) In the information processing device **10**, in a case where the conference progress prediction unit **115** predicts that the currently ongoing conference does not correspond to the "successful conference" (Step S**404**; No), that is, in a case where it is predicted that the conference corresponds to the "unsuccessful conference", it is determined which pattern of the "unsuccessful conference" the currently ongoing conference is classified into (Step S**404**). In a case where the problem resolution level prediction unit **1152** determines that the currently ongoing conference corresponds to the conference of the pattern A (Step S**404**; pattern A), the proposal unit **116** causes the output unit **140** to display, for example, "What kind of opinion do you have, Mr. OO?" in such a manner as to encourage a user whose second satisfaction level calculated by the second satisfaction level calculation unit **302** included in the biological information processing unit **300** is 25 or higher

(positive) to make a statement. In a case where the conference progress prediction unit **115** determines that the currently ongoing conference corresponds to the conference of the pattern B (Step S**404**; pattern B), the proposal unit **116** causes the output unit **140** to display a "topic change" (Step S**408**). In a case where the conference progress prediction unit **115** determines that the currently ongoing conference corresponds to the conference of the pattern C (Step S**404**; pattern C), the proposal unit **116** causes the output unit **140** to display information encouraging a user whose second satisfaction level based on the biological information and fourth satisfaction level based on the classification information are not consistent to make a statement (such as "Are you sure to agree with this opinion, Mr. OO?") (Step S**410**). In a case where it is determined that the currently ongoing conference corresponds to the conference of the pattern D, the proposal unit **116** extracts a "problem proposal"-assigned statement similar to a "problem proposal"-assigned statement in the currently ongoing conference from the minutes database unit **122** (Step S**412**). When the "problem proposal"-assigned statement similar to the "problem proposal"-assigned statement is extracted, the proposal unit **116** extracts a "problem handling"-assigned statement corresponding to the "problem proposal"-assigned statement from the minutes database unit **122** (Step S**414**). The proposal unit **116** ranks the extracted "problem handling"-assigned statements in descending order of the number of times of assignment of the classification information of the "satisfaction level" and the "problem resolution level" (Step S**416**). When the "problem handling"-assigned statements are ranked, the proposal unit **116** displays the "problem handling"-assigned statements on the output unit **140** in order of the ranks (Step S**418**). As a result, it is possible to support the currently ongoing conference to be successful.

(137) FIG. **13** is a description diagram for describing the function of the proposal unit. The function of the proposal unit **116** will be described in detail with reference to FIG. **13**. When the first user participating in the conference states "There is not specifically a problem", text data **50**A**4** of the statement is output to the output unit **140** together with the mark **40**A of the first user.

(138) In the information processing device **10**, the conference progress prediction unit **115** determines a status of the conference on the basis of the user information of the currently ongoing conference, and the proposal unit **116** extracts minutes data **1221** including a "problem proposal"-assigned statement similar to the "problem proposal"-assigned statement in the currently ongoing conference in a case where the conference progress prediction unit **115** infers that the currently ongoing conference corresponds to the "unsuccessful conference" although discussion is about to be settled. The proposal unit **116** extracts, from the extracted minutes data **1221**, a related "problem proposal"-assigned statement although the "problem" is not discussed in the currently ongoing conference. The proposal unit **116** extracts, from the extracted minutes data **1221**, a "problem handling"-assigned statement corresponding to the "problem proposal"-assigned statement.

(139) When the extraction is completed, the proposal unit **116** causes text data **50**E**2** "A problem of XX is likely to be generated in the progress of the conference in the future. Countermeasures are as follows" to be displayed together with the mark **43**. In addition, the proposal unit **116** respectively outputs, to the output unit **140**, the extracted related "problem proposal"-assigned statement as a proposal **60**C and the "problem handling"-assigned statement corresponding to the related "problem proposal"-assigned statement as a proposal **60**D.

(140) When the first user checks the proposal **60**C and the proposal **60**D displayed on the output unit **140** and states "Oh, the problem of XX may be generated. YYYY from the System may be helpful", text data **50**A**5** of the statement is output to the output unit **140**. Then, when another user participating in the conference states "We did not assume that XX would be generated", the output unit **140** is caused to output text data **50**D**3** of the statement together with a mark **40**D of another user.

(141) After a statement of a user A, when the users participating in the conference determine that the proposal **60**D has been helpful for problem resolution and assign the classification information of the "problem resolution level" thereto, a proposal **60**D**1** is displayed after the text data **50**A**5**,

and the mark **70**C indicating the "problem resolution level" is displayed. With respect to the proposal **60**D**1**, the users participating in the conference cause the mark **70**D indicating the "satisfaction level" to be displayed below the proposal **60**D**1** after the statement of the user A. The information processing device **10** stores, into the minutes database unit **122**, the classification information input with respect to the proposals during the conference. As a result, usage for learning in the second learning unit **1132** is possible.
(142) A method of preparing past minutes data of a case where minutes of a conference are recorded on paper will be described. Paper minutes are scanned, and data converted into portable document format (PDF) data is subjected to optical character recognition (OCR) and converted into text data. Then, the minutes data **1221** may be created by assignment of classification information to the text data of each of statements by the users.
(143) Furthermore, in a case where minutes of a conference are recorded as electronic data that mainly includes text data of statements of the users and does not include voice information, biological information, image information, and classification information, the information processing device **10** may directly assign the classification information to the text data by using the automatic classification unit **114**, which includes the first learned model generated by the learning unit **113**, with respect to the electronic data including the text data.
(144) The information processing device **10** may be a GPU server including a plurality of graphics processing units (GPU) with respect to various kinds of arithmetic processing. The GPUs are processors suitable for performing typical and enormous calculation processing. By using the GPU server, it is possible to efficiently perform learning in the learning unit **113**.
(145) Although the first learning unit **1131** performs learning with the satisfaction level being included as the classification information and assigns the satisfaction level with respect to the statement during the conference as the classification information in the above embodiment, this is not a limitation. It is only necessary for the first learning unit **1131** to detect at least whether a statement of a user is a problem proposal or problem handling. As a result, the conference support system **1** can check when the problem proposal and the problem handling are made in the statement during the conference.
(146) Note that the present technology can also have the following configurations.
(147) (1)
(148) A conference support system including: a user information acquisition unit that acquires voice information of each of users; and an analysis unit that generates text data of each user statement on a basis of the voice information, wherein the user information acquisition unit acquires classification information of a problem proposal or problem handling assigned by the users to the text data of the user statement, and acquires classification information of a satisfaction level or a problem resolution level assigned by the users to the text data to which the classification information of the problem proposal or the problem handling is assigned.
(2)
(149) The conference support system according to (1), further including a learning unit that generates a first learned model in which a relationship between the text data of the user statement and the classification information is learned, and an automatic classification unit that inputs the text data of the user statement to the first learned model, infers the classification information assigned to the text data of the user statement, and assigns information of the problem proposal or the problem handling to the text data of the user statement in a conference body.
(3)
(150) The conference support system according to (2), wherein the analysis unit calculates satisfaction levels of the users participating in a conference on a basis of user information including voice information, biological information, or image information of the users.
(4)
(151) The conference support system according to (3), wherein the analysis unit calculates a

problem resolution level prediction value of a currently ongoing conference on a basis of the user information.
(5)
(152) The conference support system according to (4), further including a minutes database unit that stores minutes data of a conference which data includes relevance information indicating whether the conference becomes successful or unsuccessful, wherein the learning unit includes a second learning unit that learns the minutes data and generates a second learned model in which the user information is an input and the relevance information is an output.
(6)
(153) The conference support system according to (5), wherein the analysis unit includes a proposal unit that processes the minutes data included in the minutes database unit by using the user information of the currently ongoing conference, and that outputs a proposal for the conference body which proposal is generated from an analysis result of the minutes data.
(7)
(154) The conference support system according to (6), wherein the proposal unit processes the minutes data included in the minutes database unit by using the problem resolution level prediction value, and outputs the proposal for the conference body.
(8)
(155) The conference support system according to (7), wherein the user information acquisition unit acquires classification information assigned by the users to the proposal of the proposal unit, and the learning unit performs relearning of the first learned model by using data to which the proposal of the proposal unit and the classification information are added.
(9)
(156) A conference support method including the steps of: acquiring voice information of a user; converting a statement of the user from the voice information into text data; acquiring classification information with respect to the text data; learning relevance between the classification information and the text data and generating a first learned model; and automatically assigning the classification information to the text data by using the first learned model.
(10)
(157) A conference support program for causing a computer to execute the steps of acquiring voice information of a user, converting a statement of the user from the voice information into text data, acquiring classification information with respect to the text data, learning relevance between the classification information and the text data and generating a first learned model, and automatically assigning the classification information to the text data by using the first learned model.
(158) Note that the effects described in the present description are merely examples and are not limitations, and there may be a different effect.
REFERENCE SIGNS LIST
(159) **10** INFORMATION PROCESSING DEVICE **100** USER INFORMATION ACQUISITION UNIT **101** VOICE INFORMATION ACQUISITION UNIT **102** BIOLOGICAL INFORMATION ACQUISITION UNIT **103** IMAGE INFORMATION ACQUISITION UNIT **104** CLASSIFICATION INFORMATION ACQUISITION UNIT **110** ANALYSIS UNIT **111** USER INFORMATION PROCESSING UNIT **112** NATURAL LANGUAGE PROCESSING UNIT **113** LEARNING UNIT **114** AUTOMATIC CLASSIFICATION UNIT **115** CONFERENCE PROGRESS PREDICTION UNIT **116** PROPOSAL UNIT **120** STORAGE UNIT **121** USER INFORMATION STORAGE UNIT **122** MINUTES DATABASE UNIT **123** CORPUS UNIT **124** DICTIONARY UNIT **130** COMMUNICATION UNIT **131** TRANSMISSION UNIT **132** RECEPTION UNIT **140** OUTPUT UNIT

Claims

1. A conference support system, comprising: a processor configured to: acquire voice information of each user of a plurality of users; generate, based on the voice information, text data of each user statement of a plurality of user statements of the plurality of users; acquire first classification information assigned by the plurality of users to the text data of each user statement of the plurality of user statements; generate a first learned model by learning relevance between the first classification information and the text data; train the first learned model, based on supervised learning, using at least one of the voice information, biological information of at least one user of the plurality of users, or image information of at least one user of the plurality of users; and automatically assign second classification information to the text data based on the trained first learned model.
2. The conference support system according to claim 1, wherein the processor is further configured to: input the text data to the first learned model; infer the second classification information assigned to the text data; and assign information of at least one of a problem proposal or a problem handling to the text data of a user statement, of the plurality of user statements, in a conference body.
3. The conference support system according to claim 2, wherein the processor is further configured to: calculate satisfaction levels of the plurality of users participating in a conference based on user information including at least one of the voice information, the biological information, or the image information.
4. The conference support system according to claim 3, wherein the processor is further configured to: calculate a problem resolution level prediction value of the conference based on the user information.
5. The conference support system according to claim 4, wherein the processor is further configured to: store minutes data of the conference, wherein the minutes data includes relevance information indicating whether the conference becomes one of successful or unsuccessful; learn the minutes data; and generate, based on the learned minutes data, a second learned model in which the user information is an input and the relevance information is an output.
6. The conference support system according to claim 5, wherein the processor is further configured to: process the minutes data based on the user information of the conference; and output a proposal for the conference body based on the minutes data.
7. The conference support system according to claim 6, wherein the processor is further configured to: process the minutes data based on the problem resolution level prediction value.
8. The conference support system according to claim 7, wherein the processor is further configured to: acquire third classification information assigned by the plurality of users to the proposal; and perform a relearning operation of the first learned model, based on data to which the proposal and the third classification information are added.
9. A conference support method, comprising: acquiring voice information of each user of a plurality of users; generating, based on the voice information, text data of statement of each user statement of a plurality of user statements of the plurality of users; acquiring first classification information assigned by the plurality of users to the text data of each user statement of the plurality of user statements; generating a first learned model by learning relevance between the first classification information and the text data; training the first learned model, based on supervised learning, using at least one of the voice information, biological information of at least one user of the plurality of users, or image information of at least one user of the plurality of users; and automatically assigning second classification information to the text data based on the first learned model.
10. A non-transitory computer-readable medium having stored thereon, computer-executable instructions which, when executed by a computer, cause the computer to execute operations, the operations comprising: acquiring voice information of each user of a plurality of users; generating, based on the voice information, text data of statement of each user statement of a plurality of user statements of the plurality of users; acquiring first classification information assigned by the

plurality of users to the text data of each user statement of the plurality of user statements; generating a first learned model by learning relevance between the first classification information and the text data; training the first learned model, based on supervised learning, using at least one of the voice information, biological information of at least one user of the plurality of users, or image information of at least one user of the plurality of users; and automatically assigning second classification information to the text data based on the first learned model.